

Predicting MLB Draft Selection, Order, and Bonus from Public NCAA Statistics

Abstract

A college baseball player deciding whether to enter the MLB Draft wants to know three things: will they be drafted, where will they go, and what will they be paid. Existing college baseball and MLB Draft modeling relies on data that cannot be readily redistributed or reproduced: the sabermetric leaderboards most college baseball work relies on are not free, so results cannot be checked by the public. The first contribution of this work is to construct and validate a public NCAA-derived dataset, consisting of 61,270 Division I player-seasons from 2021 through 2026 (every player with any playing time), of whom 2,565 (4.19%) were drafted.

We then use MLB Draft prediction as a substantive benchmark of what can, and cannot, be inferred from public performance data alone. The headline test is a 2026 draft board built with no hindsight: every model retrained on 2021–2025 alone, every field that exists only after a player is drafted removed, and all 10,434 2026 player-seasons ranked from scratch. Of the top 50 names, 48 were actually drafted, matching MLB Pipeline’s college-only top 50. Our board runs 261 deep and is still 71% accurate at that depth, where the public scouting board stops at 142 college players.

This paper contributes both an open dataset that matches batter and pitcher data of a paid source at 99% and a predictive pipeline demonstrating that data’s utility in modeling if a player is drafted, where they will be drafted, and their signing bonus. We find that public college-level data is not a replacement for expert opinions.

1 Introduction

The MLB Draft is held every July and assigns roughly 600 amateur players to professional organizations across 20 rounds [2]. For a college player the decision to enter the pros is close to irreversible. Sign, and your college eligibility is gone. Go back to school, and you risk injury, a worse season, or a deeper draft class next year. The players with the most at stake are juniors, who still hold a year of eligibility and can use it as leverage.

Public prospect rankings exist [10, 19, 20], but they cover a few hundred names out of a Division I population above ten thousand, and they do not tell a player which of their own numbers is helping or hurting them.

College baseball analytics also leans on a few sabermetric leaderboards that are not free to redistribute [8, 11]. Since most of it is locked up behind paywalls, it makes published results awkward to verify and to build on. A major gap we aim to address is a single public repository of NCAA data that can consistently provide these measures, and be used for further analysis.

Several studies have explored MLB Draft results previously. Karcher [15] showed that, across the 1996–2011 drafts, two thirds of signed first-round picks reached the majors, falling under 10% by round 20. Chandler and Rosenbaum [5] found that about 39% of top-50 picks never reach the majors at all. Penn [23] compares college and high-school draftees taken at the same slot and finds a clear college advantage in the first six rounds. These papers are largely concerned with modeling outcomes post-draft, and other works do the same [1, 7, 13, 26].

There are fewer papers which attempt to model the full MLB Draft pipeline (selection-placement-bonus) itself. Garmon [12] estimates two linked equations on the top 300 picks from 2003 to 2010: whether team and player agree at all, and what bonus they settle on. That two-part structure is the closest published relative of our own draft-probability and bonus models. Pifer et al. [24] turns career-state probabilities into a dollar threshold for whether a draftee should sign.

College statistics are hard to compare across programs because schedules differ enormously, and Kilanowski and Moloney [16] addresses this inside the Cape Cod League, where the competition is level. A companion student study [17] struggled to produce meaningful predictions: every player projected landed between the 100th and 500th selection. In Danovitch [9], written scouting reports were directly used with deep learning methods to predict how well an amateur baseball player will make it to the major leagues. The models showed moderate success, with the TextCNN model achieving the best overall results with 69.02% accuracy and a 56.42% F-1 score.

This paper makes two major contributions:

1. **Data/resource contribution:** A re-distributable NCAA Division I dataset of 61,270 player-seasons built from public NCAA statistics, with data processing code and reconstructed sabermetric features. Quantitatively validated against a proprietary source to establish agreement.
2. **Benchmark contribution:** a set of models for the full MLB Draft pipeline, for draft selection, ordering, and compensation prediction. Demonstrating the capabilities and limitations of public NCAA data.

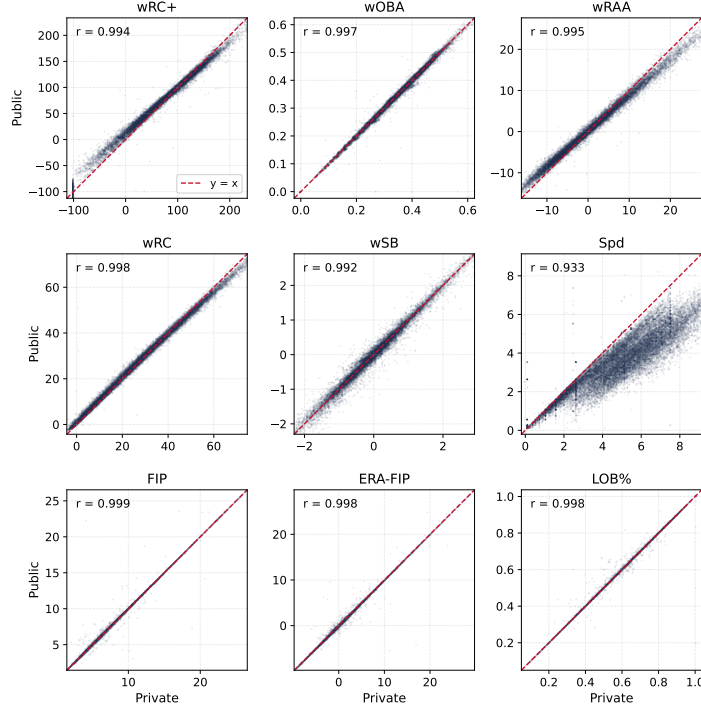


Figure 1: Hexbin panels that compare the nine sabermetric columns that were recomputed from public statistics to the private dataset.

The paper proceeds as follows: in Section 2, we describe the process of building the public dataset, validating against a pay-walled equivalent, and the features and targets we utilize in the machine learning pipeline. Section 3 briefly describes the methods used for building the predictive models. Section 4 covers the results and Section 5 the discussion.

2 Data

2.1 Building an Open Dataset

Our player statistics come from public NCAA data rather than a paid leaderboard [22], and we checked the substitution column by column against the paid version. Of 26,821 batting seasons, we matched the same player in the paid file 99.4% of the time, and 94.3% of the raw counts (hits, walks, home runs) were identical. Pitching matched 99.8% of players with 78.2% of counts identical. Every rate built by pure arithmetic (batting average, on-base percentage, slugging, ISO, walk and strikeout rate, BABIP, ERA, WHIP, K/9) agrees exactly wherever its inputs agree.

This column-level check covers 2021–2025; 2026 arrived separately and is checked only by our own acceptance tests. Nine columns are league-relative rather than raw performance statistics: wOBA, wRC, wRC+, wRAA, wSB, Speed Score, FIP, ERA minus FIP, and strand rate (LOB%). These depend on run values fitted to a league, so they cannot be copied directly. We fit our own constants from public data and recompute them.

Our sabermetric columns track the paid versions closely (wRC+ $r = 0.9915$, wOBA $r = 0.9950$, wRAA $r = 0.9928$, wRC $r = 0.9980$, wSB $r = 0.9900$, Speed Score $r = 0.9326$, FIP $r = 0.9985$, ERA minus FIP $r = 0.9975$, LOB% $r = 0.9978$) as shown in Figure 1.

The public source we use for 2021–2025 stopped updating midway through the 2026 season, holding only about 60% of the year. Rather than publish understated 2026 numbers we scraped all 308 Division I team-seasons ourselves. Sources for 2021–2024 are pinned by upstream commit and checksum; 2025 and 2026 come from local scrape caches, which we ship as derived files because no public mirror covers them.

Our paid source publishes a player’s date of birth only for players it has already linked to a professional record. On that paid population, age was missing for roughly 43% of college player-seasons but for only 0.3% of the ones that ended in a draft selection. Public NCAA data has no birth date at all, instead the class year is available for 99.9% of players.

In addition to performance, inputs also include twelve columns describing the athletic budget of a player’s program, motivated by Caro and Elder [4], who found that spending explained roughly 18% of the variance in winning among Power Five baseball programs. If money buys wins and a harder schedule, it may also buy the attention of scouts. Throughout, we report precision-recall curves rather than ROC curves, because drafted players are rare and ROC curves flatter a model

on rare events [25].

We also developed an open source python package which includes all the data necessary to replicate this work, as well as other functions including a way to generate scouting reports using the models. We then developed an interactive public website where users can browse the board, look up a player, or score a stat line of their own. For the purposes of this competition, we have omitted the details of the package and site for identification purposes, but can produce it as necessary.

2.2 Population, targets, and features

The unit of observation is the player-season. We keep every player with any playing time, rather than only those clearing a qualification threshold, because a player deciding whether to enter the draft does not get to know in advance whether they will qualify. That triples the player pool and drops the draft rate by more than half, from 9.4% among qualified players to 4.2% (Table 1).

Year	Player-seasons	Drafted	Rate
2021	9,817	411	4.19%
2022	10,039	417	4.15%
2023	10,092	420	4.16%
2024	10,393	435	4.19%
2025	10,495	426	4.06%
2026	10,434	456	4.37%
Total	61,270	2,565	4.19%

Table 1: Player-seasons and draft selections by year. The rate is stable near 4%, so no single season drives the results.

Draft results and signing bonuses come from MLB’s public statistics API [18]. Joining on year and pick number attached a draft record to all 2,565 drafted player-seasons; a name-similarity screen then removed one row where two similarly named players had been conflated, leaving 2,564.

There are three prediction targets. The first is whether a player was drafted. The second is their position in the college draft order: the rank of a drafted player among college draftees that year, running from 1 to about 455, with a version rescaled to run from 0 to 1 so that years of different sizes are comparable. Finally, we consider answering the final question of predicting the signing bonus.

We model signing bonus in percentage terms rather than raw dollars, because the distribution is right-skewed. Bonuses run from \$500 to \$10,350,000 with a median of \$150,000, so being off by \$50,000 is far worse at the bottom of the draft than the top. Bonuses are recorded for 2,494 of 2,564 drafted players (97.3%).

The models see 155 inputs: 70 describing the player, 52 their team’s season, 14 team RPI and schedule strength, 7 playing-time shares, and 12 the athletic budget of their program. The playing-time shares exist because the raw data keeps a player’s totals and their team’s totals in separate columns; they are what separate a starter from a reliever with identical rate statistics, or an everyday player from a part-timer. The program-finance columns come from the federal Equity in Athletics Disclosure Act survey [27], the only block not derived from on-field results. We include both dollar figures and within-year percentile ranks, because budgets inflate a few percent every year while a program’s budget *rank* barely moves.

In place of age we use three columns: class year, how many calendar years the player has appeared in our data, and the class year they first appeared at. Together they capture most of what age was doing: how far along a player is, and whether they have been around longer than their class implies.

The models trained on drafted players also see five fields describing a player physically: height, weight, primary position, bat side, and throwing hand. These come from MLB’s draft records, so they exist only for players who were drafted.

Missing values are left missing. Roughly half of all rows have no ERA, because half of all players are not pitchers, and 98% have no team triple plays, because triple plays almost never happen. Our chosen methodology, XGBoost, naturally handles missing data.

3 Methods

3.1 Model Specification

All models are implementations of XGBoost [6] using decision tree-based architectures. The general form of a boosted-tree predictor is, for player i with features X_i :

$$F(X_i) = F_0 + \sum_{m=1}^M \eta f_m(X_i),$$

where f_m is the m -th regression tree, M is the number of boosting iterations, and η is the learning rate.

The feature vector X_i is a single season-level record for player i : 155 columns, all observable before the draft, in five blocks. **Player performance (70)**: pitching counting totals and rates (ERA, FIP, WHIP, K/9, BB/9, K-BB%), batting equivalents including wRC+, wOBA, and ISO, plus role and class standing. **Team performance (52)**: the player’s program that season with their record, Pythagorean expectation, and full team batting, pitching, and fielding lines. **Schedule strength (14)**: RPI, non-conference RPI, SOS, conference rank, and winning percentage split by venue and by opponent quadrant. **Playing time (7)**: ratios formed so the trees need not learn the division themselves, including share of team innings and at-bats. **Program resources (12)**: EADA budget columns in log dollars and within-season percentile. This is the only block not derived from on-field results. The draft-order and bonus models add five attributes recorded at signing (height, weight, position, bats, throws); the pre-draft simulation drops these so the full population can be scored.

For the draft-selection classifier, the XGBoost utilizes the binary logistic objective, and draft probability is estimated by:

$$\hat{p}_i = \sigma(F(X_i)) = \frac{1}{1 + \exp(-F(X_i))}$$

For the continuous-outcome models of draft order and bonus, the identity link function is used for predictions and the mean squared error objective function is used.

For the draft-selection classifier, we evaluate performance with two main measures. *Precision* is: of the players we flag, what share were really drafted. *Recall* is: of the players who were really drafted, what share did we flag. The area under the precision-recall curve, or PR-AUC, summarizes the trade-off between them across every possible cutoff, and it must be read against the base rate: a model that knows nothing about the player scores about 0.042 here, because that is how often a randomly chosen player is drafted. For making final predictions in the classifier, we chose a cutoff of $\hat{p}_i$ of 0.4.

For the draft order model, we use *Spearman correlation*, which asks whether we put players in roughly the right order regardless of exact position, alongside the average number of places a player is missed by.

In the signing bonus model, we report the typical regression error and the evaluation measures R^2 .

3.2 Prospective 2026 Evaluation

Everything above still assumes we know who was drafted, which is exactly what a player in May/June does not know. So we rebuild all models under two restrictions: train on 2021–2025 only, so 2026 is entirely unseen; drop the five physical fields that exist only for drafted players, so every player in the population can be scored. We then score all 10,434 2026 player-seasons and build a board in three steps: keep the players, the model gives better-than-even odds, drop players who are not yet draft-eligible, and then sort by predicted order.

The 40% cutoff was chosen via hyperparameter tuning as it struck the best balance of maximizing F1 score while penalizing False Positive rate, as we assume that falsely informing a player they are likely to be drafted is a worse error than falsely informing them they will not be.

The separate Stage 1 draft-probability model uses a stratified hold-out that is split at random rather than by year, and is not grouped by player, so one of a player’s seasons can sit in training while another sits in the hold-out; the leave-one-season-out figures are free of both and we report them alongside. The order and money models pick how many question trees to build by watching their own 2026 test set, which makes their numbers mildly optimistic. The board simulation does none of this, and is why we report it first.

4 Results

4.1 The 2026 draft board

We lead with the board because it is the only result with no 2026 information anywhere in training.

Scoring all 10,434 2026 player-seasons, the retrained model reaches a PR-AUC of 0.614 against a base rate of 0.044 (Figure 2). At the 40% cutoff it flags 513 players, of whom 252 were really drafted, giving precision 0.635 and recall 0.491. Opening the gate wider trades precision for reach; \$5.2 gives the exchange rate.

The eligibility filter is where the loss of a birth date hurts the model, and the shape of the loss is instructive. Draft eligibility for a four-year college player is a disjunction: they qualify either by having completed their junior year or by having turned 21 [3]. Public NCAA data observes the first term exactly, since class year is published for 99.9% of players,

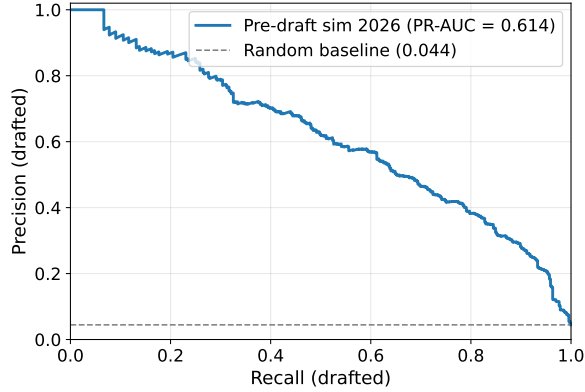


Figure 2: Precision-recall curve for the 2026 board, scoring all 10,434 player-seasons. Every model was trained on 2021–2025 only, with post-draft fields removed and nothing set aside to check against, so no 2026 information reaches training. The area under this curve is 0.614 against the 0.044 a model that knows nothing about the player would get.

K	Our board	MLB Pipeline
50	48/50 (96%)	48/50 (96%)
100	88/100 (88%)	96/100 (96%)
142	118/142 (83%)	134/142 (94%)
200	158/200 (79%)	n/a
261	186/261 (71%)	n/a

Table 2: Board quality: of the top K names, how many were actually drafted, a test of selection only, so a player counts as a hit wherever they were taken. MLB Pipeline’s published top 250 contains 108 players our dataset cannot hold, predominantly high-schoolers, so we compare against its college subset, which exhausts at 142. $K = 142$ is the deepest like-for-like comparison; our board ends at 261.

and the second term not at all. Our filter can therefore implement only one branch of a two-branch rule, and every player it gets wrong is one who qualified through the branch we cannot see.

For example, the filter cut 17 flagged players and 6 of them, 35%, were drafted anyway, including the 32nd and 46th overall picks. All six are sophomores with two counted seasons: ineligible under the only branch we can evaluate, yet evidently eligible in fact, and for a four-year college player age is the only other route the rule allows. A 21-year-old sophomore and a 19-year-old sophomore are identical on every public field.

Table 2 scores the board against reality and against MLB Pipeline’s published top-250 prospect list [20]. 108 of Pipeline’s 250 ranked prospects are players our dataset cannot contain: predominantly high-schoolers, plus junior-college and non-Division I players and any name we could not match. Counting those as misses would penalize Pipeline for covering a population we deliberately excluded, so we score it on its college subset, which runs out after 142 names. Note also what this table does and does not measure: a name counts as a hit if the player was drafted at all, wherever they were taken, so this is a test of selection and not of order.

Pipeline’s rankings are not an independent benchmark in the way a truly blind prediction would be. Front offices, agents, and even players see these rankings before draft day. MLB’s own draft tracker lists Pipeline rank alongside each player’s school and draft slot [21], so the ranking is plausibly part of the information teams are drafting on, not just a forecast of it. None of this is a knock on Pipeline’s scouting, since it’s a well-earned, well-informed list. But it means Pipeline’s ordering accuracy benefits from a degree of self-fulfillment that our model, working purely off performance and physical data, cannot draw on.

We tie the scouting consensus at the top and lose to it below: 48 of 50 against 48, 88 of 100 against 96, and 118 of 142 against 134 at the deepest depth both lists reach. Pipeline ranks 142 college players in 2026, our board ranks 261 and still hits 71% at that depth. Both lists are near the ceiling of this test at $K = 50$, since “was drafted at all” is an easy bar for a top-50 prospect, so the tie there means less than it looks, though our board reaches it despite the eligibility filter having discarded the 32nd and 46th overall picks.

The other models hold up on the board. Among the 185 board players who were drafted, the order model reaches a Spearman correlation of 0.655 and places a typical player about 35 spots from their real position, under two rounds; Pipeline reaches 0.818 on the 98 it also ranks, beating us here as it does everywhere on ordering. The bonus model reaches 0.715 across 181 players with a typical error of \$590,166, larger than the median bonus itself, which is the clearest sign

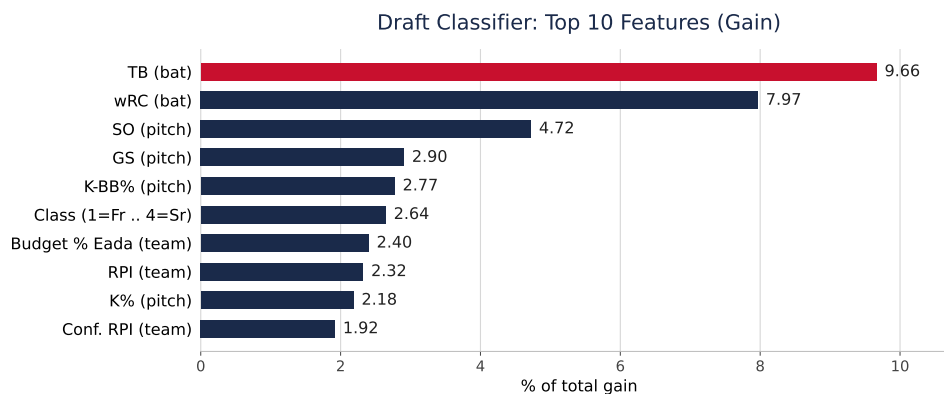


Figure 3: How much the top 10 features contribute to a prediction for the draft-probability model. Offensive volume leads (total bases, runs created), followed by pitcher strikeouts and starts, class year, the program’s athletic-budget percentile and team RPI. Bars show how much each statistic moved the model’s projections.

its dollar figures are a ranking rather than a quote.

Two players show the shape of the remaining error. The actual first overall pick, Roch Cholowsky, came eighth on our board. The player we ranked first, Hunter Dietz, went 22nd overall, still the first round but not the top. Selection is where a statistical model helps. Ordering is where it does not.

4.2 Will they be drafted?

On a stratified hold-out of 12,254 players at the true 4.19% draft rate, the model returns a PR-AUC of 0.608 against the 0.042 a model that knows nothing would get, a 14.5-fold improvement.

We do not lead with accuracy, even though ours is 96.8%. A model that answered “no” to every player would score 95.8%, because only 4.19% of player-seasons end in a selection, so ours beats it by less than a point (96.4%). What separates them is invisible in the accuracy column: the do-nothing model finds none of the 513 drafted players and ours finds 252. The same asymmetry runs through the table. The model recalls 98.8% of the undrafted, where guessing “not drafted” is nearly always right, and 49.1% of the drafted. The weighted averages inherit that imbalance and flatter the model, because 96% of the rows are the easy class. Only the drafted row describes the task anyone cares about.

Refitting six times and leaving one season out each time gives 0.597 pooled over all 61,270 player-seasons, ranging from 0.551 in 2022 to 0.625 in 2025. The two agree closely, and since the hold-out is split at random across years and the leave-one-season-out figure is not, that agreement says the random split is not doing much work here.

At a gate of 0.30, the model catches 54.4% of draftees at 56.5% precision; at 0.20, 64.1% at 48.7%. A player weighing their own decision wants the wide end, a published board the narrow one, which is why the board gates at 0.40.

Figure 3 ranks what the model leans on. Figure 4 shows how some features move the model based on their inputs, and how some features interact with others. Raw offensive volume leads: total bases first, then runs created, then pitcher strikeouts and pitcher starts. Class year is sixth, the program’s athletic-budget percentile seventh and team RPI eighth. The model rewards players who produced a lot, at good programs, on hard schedules, roughly what a scout would do. Potentially, with some of the same institutional bias baked in.

4.3 Where in the college order?

Across the 455-player 2026 class, the order model gets the broad shape of the draft right, with the early names mostly early and the late names mostly late, but misses an individual player’s slot by about 88 places, roughly three rounds. That compares against 114 places for predicting every player at the class average, and the model accounts for 39% of the spread in actual order. The rank correlation is 0.630 (95% interval 0.571 to 0.683). The rescaled version of the target performs essentially identically at 0.621, so nothing hinges on that choice. Figure 5 shows the whole class beside a top-50 zoom.

Figure 6 shows the residuals, the predicted minus actual college draft order. The errors fan out with the prediction: players we place early are placed roughly right, while players we place late are missed by a hundred places either way. The top of our board is worth reading closely; the bottom is worth reading as a set rather than an order.

Table 3 runs the same comparison across all six years. For each year we train on the other five and predict the held-out one, then compare against MLB Pipeline on the players both rankings cover.

Two things stand out. Our year-to-year correlation ranges from 0.383 to 0.694 with no trend, while Pipeline’s stays between 0.762 and 0.857; that instability is in our model, not in the draft, and Pipeline’s steadiness over the same six drafts is the evidence for it. Second, the mean gap of 0.26, about ten places within the roughly 120 players both rankings cover,

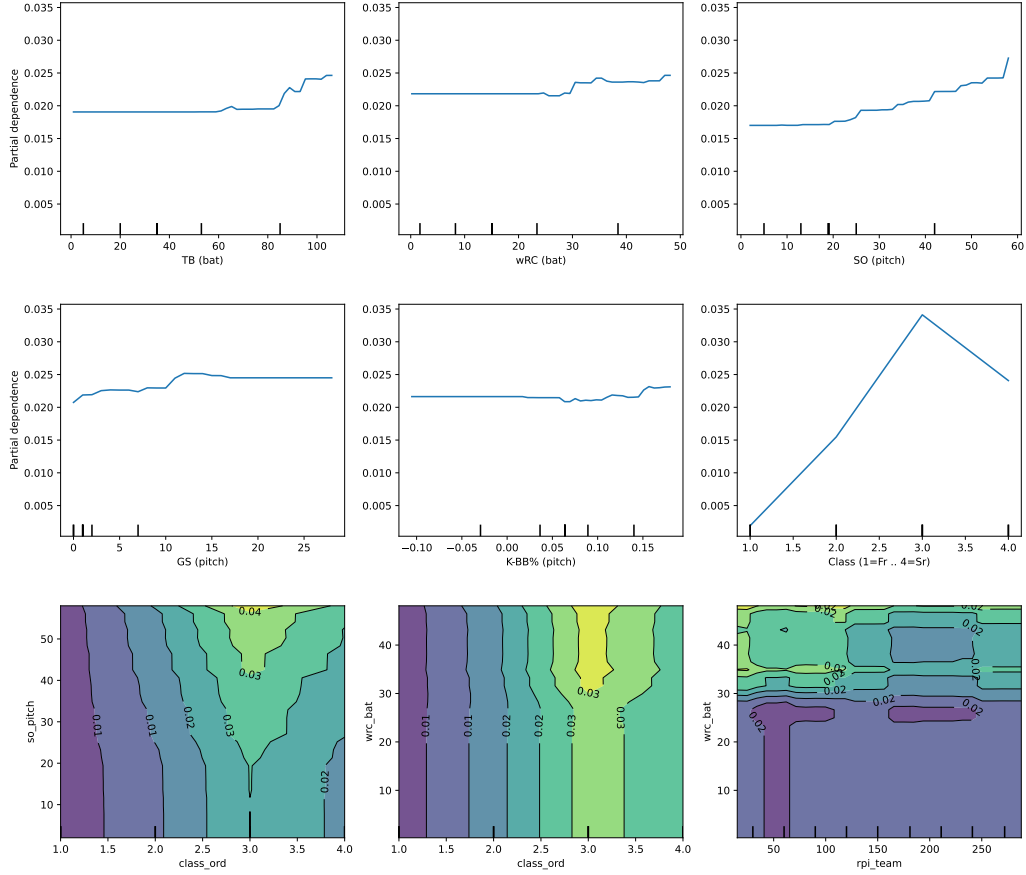


Figure 4: Taking some of the features from Figure 3, we can map out how they interact with the model and each other.

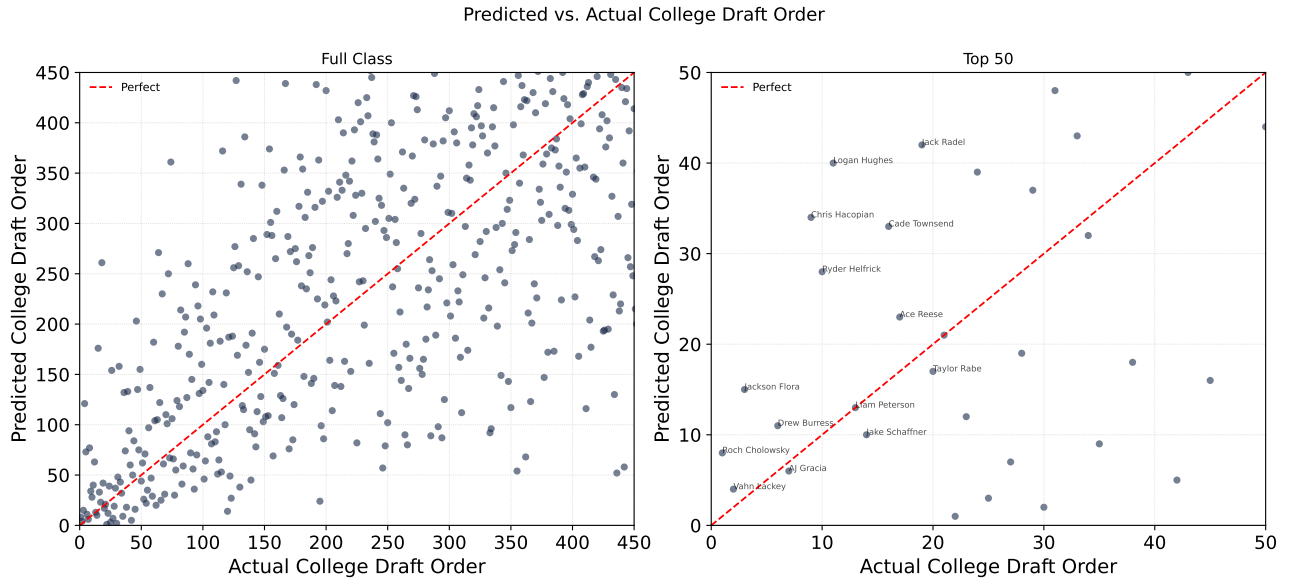


Figure 5: Predicted against actual college draft order for the held-out 2026 class ($n = 455$). Points on the dashed line are exactly right. On the left, the whole class: the cloud tilts the right way but is wide, and predictions compress toward the middle, so the deepest picks are systematically projected earlier than they go. On the right, the top 50 with the leading names labelled. Agreement is loose even here, though the eventual first two picks, Roch Cholowsky and Vahn Lackey, both land inside our top ten.

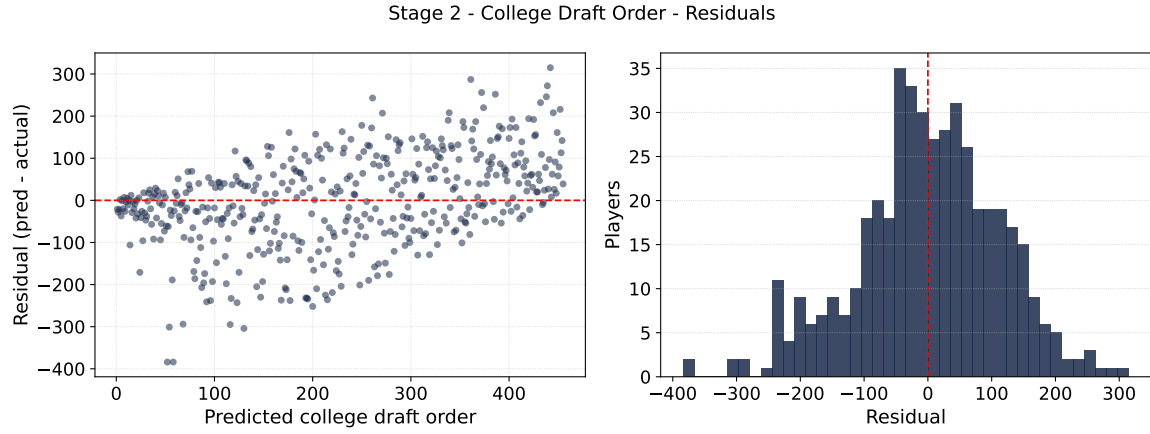


Figure 6: Stage 2 residuals on the held-out 2026 class, predicted minus actual college draft order. Left, against the prediction: the spread widens steadily, so a projection of 40th is far more trustworthy than a projection of 400th. Right, the distribution: centred near zero with a long tail both ways, and a typical miss of 88 places.

Year	n	Ours ρ	Pipeline ρ	Ours places	Pipe. places
2021	115	0.383	0.817	28.8	15.2
2022	121	0.530	0.857	26.1	14.0
2023	132	0.616	0.762	25.2	19.0
2024	116	0.694	0.832	20.0	14.4
2025	124	0.642	0.838	23.5	14.6
2026	134	0.481	0.804	31.0	17.3
Mean	(742 total)	0.558	0.818	25.8	15.8

Table 3: Draft-order accuracy against MLB Pipeline, on the college players both rankings cover, 2021–2026. Our model swings from year to year; Pipeline does not. Places-missed figures are within each year’s matched pool of roughly 120 players, not out of the full class, so they are not comparable to the 88 places in the text.

is a rough price tag on what in-person scouting adds to public statistics. Part is substantive: scouts see pitch movement, physical projection and medical history. Part is structural, because pick number also reflects bonus demands, signability, team strategy and positional need, none of which appear in a box score.

4.4 A second scorecard: who actually made it

Draft order is a flawed thing to be graded against, for the reason just given. So we also scored each ranking against something the draft cannot contaminate: the professional level each player eventually reached. For the 2021–2023 cohorts we compare our rankings, Pipeline’s, and the actual draft order against how far each player climbed in the MiLB and MLB. That outcome is never an input to any of our models.

A limit to this scorecard is that the data is restricted to college players MLB Pipeline already ranked in its top 250, so “our top 50” is the top 50 of about 120 players somebody else already selected, and that pool is strong to begin with: about 79% of it reached Double-A or higher however it was ranked.

On ordering we trailed. We ranked players in the order they would eventually climb the minor-league ladder about half as well as Pipeline did. On membership the three are indistinguishable. Of each ranking’s top 50, the share reaching Double-A or higher is 86.0% for us, 88.7% for Pipeline and 87.3% for the actual draft, all within about eight points of the 79% that picking 50 at random from the pool would give.

4.5 What will they be paid?

The bonus model does a good job saying who will be paid more than whom, at a correlation of 0.779. When it comes to quoting how many dollars, it starts having issues and barely beats quoting the median bonus to every player. It has a typical error of \$349,940. It predicted \$957,940 for the first overall pick, who signed for \$10,350,000. Because we model bonuses in percentage terms the model aims at the typical player, and the typical drafted player signs for \$150,000. Rank players by expected earnings with this model; do not quote a first-round figure from it.

5 Discussion

A college junior weighing the draft against another year of school gets a rough joint picture: whether they are likely to be drafted, roughly where, roughly what they would be paid, and a breakdown of which of their own numbers help and which hurt. That last part is why we use a model we can pull apart, and Figure 7 shows what a player receives.

The example is also the paper’s most interesting error. Ryan Gerety, a Northeastern University junior (in the CAA), hit .326/.456/.492 (AVG/OBP/SLG) across 61 games in 2026 with 51 walks and a walk-to-strikeout ratio above one. What drives his projection is volume and standing: total bases is worth eleven percentage points, then class year and plate appearances per game at ten each, with runs created and his program’s budget behind them. Nothing in his profile counts meaningfully against him. The model puts him at 0.762 and the board at 123rd. He went undrafted, then signed with the San Diego Padres as a free agent [14].

Gerety is one of the 75 names on our board that were not drafted; we audited only him and cannot say how many of the other 74 signed somewhere, so this illustrates rather than corrects the board’s precision. But its misses should not all be read as the model failing to see what scouts saw.

For a team or an agency the board itself is the deliverable rather than any single prediction. It ranks 261 eligible players from public data, well past the depth of the public scouting board, holds 186 of 261 precision at that depth, and can be rebuilt the moment new statistics land. It also provides insights into which features have the greatest impact and which could be improved upon.

For an analytics group, Table 3 prices what in-person scouting adds over public statistics at about 0.26 of rank correlation, which is useful when splitting effort between scouts and models.

Losing the paid dataset cost us a birth date and gained us a model that is not quietly reading its own label. On balance that is a good trade, and the class-year columns carry most of what age was doing for the models themselves.

The eligibility rule is based specifically on date of birth, which is not available in our dataset.. This is an identification data problem. The consequence is the false removals of \$5.1: six players, including the 32nd and 46th overall picks. A remedy is to have a data source that has date of birth publicly available.

5.1 Limitations and Future Work

The inputs are season totals from a box score: no velocity, no exit velocity, no spin rate, no defensive measurement, no medical information. They also cannot see an arc, because each player-season is scored on its own, so a third straight strong season at a top program reads like a first one. Danovitch [9] shows much of that missing signal is written down in scouting reports and can be modeled from text.

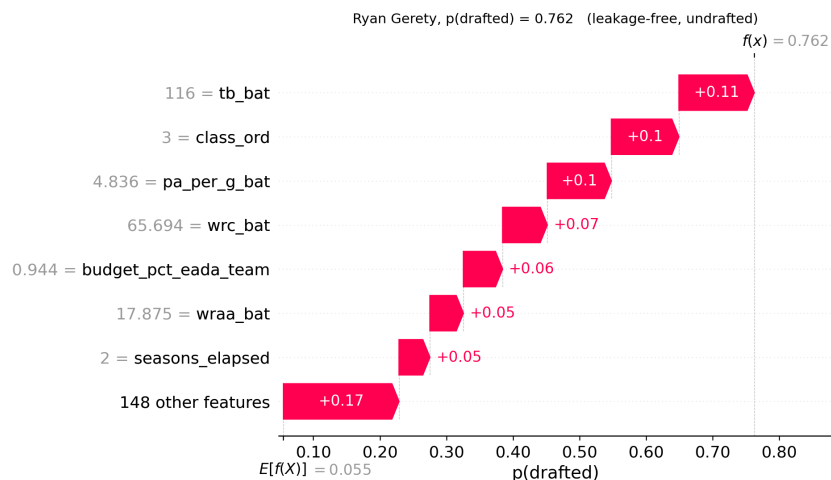


Figure 7: What one player gets back. Each bar is a statistic’s contribution to Ryan Gerety’s predicted draft probability, starting from the 5.5% a player drawn at random would get and ending at his 76.2%. The bars are probabilities, so they read directly: total bases is worth eleven points to him, class year ten. Scored by the leakage-free model of \$4.3, which never saw 2026. He was not drafted.

- The order and money models choose how many trees to build by watching their own test year, so their numbers are mildly optimistic; the draft-probability model’s hold-out is split at random rather than by year and is not grouped by player, though its leave-one-season-out figure is not. The board does none of this, which is why we lead with it.
- Six seasons is a short window, so need more years to compare year-to-year effects.
- The population is NCAA players only, and season statistics include games played shortly before the draft, so a player whose postseason moved their stock has that performance inside their totals.
- For calculating how each player climbs in MiLB, we need about 10 years more worth of data. It can potentially be linked to the Gerber [13] model.
- Trying a different modeling framework besides XGBoost, or some sort of more specifically focused ranking model for the draft order prediction.

5.2 Conclusions

We built three models predicting whether an NCAA baseball player is drafted, where they go in the college order, and what they can be paid, from 61,270 player-seasons across 2021–2026 and no paywalled data.

The headline result is a draft board with no hindsight in it. Retraining every model on 2021–2025 alone, removing every field that only exists after a player is drafted and ranking all 10,434 2026 player-seasons, 48 of our top 50 and 186 of our top 261 were actually drafted. That ties MLB Pipeline’s college-only board at the top 50 and trails it at the top 100, 88 against 96. We do not outperform professional scouts at the top; we match them there and go deeper.

Asked who will be drafted, the models return under 300 names out of ten thousand, fourteen times better than guessing. Asked where those players go, the model misses a typical player by three rounds. Asked what they will be paid, the model ranks earnings well and misquote first-round dollars.

We made our data and results publicly accessible through an open-source Python package and interactive website, allowing users to explore the draft board, look up individual players, and evaluate their own statistical profiles.

Data and code availability, and use of AI tools

Everything needed to reproduce these results is public. The paper’s repository holds the analysis notebook with its outputs, the data, the build scripts, and a table mapping every number here to the cell that produced it: github.com/---. The data package behind it is at github.com/---. Two sets of figures come from outside the notebook: the validation percentages in §3.1 and the energy figures in §6.5.

Claude Opus 5 assisted with editing, debugging, data visualizations, and the interactive site at ---.connect.posit.cloud. The models, data pipeline and analysis are the authors’ own.

Removed links to be anonymised for review.

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